

1 **Analytical solution techniques**

The following problem sets are concerned with the analytical techniques we considered in part I of TMA4125. For each of the problems sets below, write down the solutions on paper. At the end **upload a single pdf** file which includes all the answers to Problem set 1 - 4. Include detailed enough derivations and/or explanations so that it becomes clear how you arrived at your answers.

Problem 1

As usual, assume that $\delta(t)$ and $u(t)$ denote the Dirac function and the Heaviside function, respectively.

Use the Laplace transform to solve the initial value problem

$y''(t) + 9y(t) = \delta(t - 1) + u(t - 1), \quad t > 0$

with initial conditions $y(0) = y'(0) = 0$.

Problem 2

Again, let u be the Heaviside function.

a) Sketch the function $-u(x + \pi) + 2u(x)$ over the interval $[-\pi, \pi]$.

b) Compute the Fourier series $f \sim a_0 + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$ for

the function $f(x) = \cos 3x - \frac{1}{2} \sin 2x - u(x + \pi) + 2u(x)$.

Problem 3

Given is the heat equation

$\partial_t u(x, t) = \partial_{xx} u(x, t), \quad 0 \leq x \leq 4,$

together with initial condition

$u(x, 0) = 2 \sin(3\pi x)$

and boundary conditions

$u(0, t) = u(4, t) = 0, \quad t > 0.$

a) Find all solutions of the heat equation which are of the form

$u(x, t) = F(x)G(t)$ and also satisfy the boundary conditions.

b) Find the solution u to the heat equation which satisfies both the boundary and the initial conditions.

Problem 4

Given is the wave equation on the real line

$\partial_{tt} u(x, t) = 16 \partial_{xx} u(x, t), \quad x \in \mathbb{R}, t > 0,$

with initial conditions

$u(x, 0) = x, \quad \partial_t u(x, 0) = \cos(x).$

Find the solution to this problem, and simplify the resulting expression as much as possible.



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Maximum marks: 50

2 Fixed-point iteration

Given the function $g(x) = \frac{e^x}{10}$. Does this function have a fixed-point in the interval $[-1, 1]$? And if so, is this fixed-point unique? Explain your answers.

Fill in your answer here

Maximum marks: 8

3 Compute interpolation polynomial.

Given the points $x_0 = 0.0, x_1 = 1.0, x_2 = 2.0, x_3 = 4.0$ together with 4 data points $f_0 = 1, f_1 = 5, f_2 = 10, f_3 = 12$. Find the polynomial of minimal degree interpolating through these points. Express it in the form

$$p_n(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0.$$

Only state the final solution, do not give a detailed derivation.

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Maximum marks: 8

5

Composite Boole's quadrature rule

Again, let $BQ[f]$ be the so-called Boole's quadrature rule which computes an approximation of the integral $I[f] = \int_a^b f(x)dx$ and satisfies an error estimate of the form

$$|BQ[f] - I[f]| = \frac{8}{945} \left(\frac{b-a}{4} \right)^7 f^{(6)}(\xi)$$

for some $\xi \in (a, b)$.

Assume that you now use Boole's quadrature rule to create a **composite** Boole's quadrature rule $CBQ[f]$. Denote by h the length of the subintervals in the composite rule, then the quadrature error $|CBQ[f] - I[f]|$ behaves like

$$|CBQ[f] - I[f]| = \mathcal{O}(h^p)$$

for some $p \in \mathbb{N}$, and p would be the **convergence order** of the composite Boole's quadrature rule.

What convergence order do you expect for the composite Boole's quadrature?

Select one alternative:

- ☐ 9
- ☐ 6
- ☐ 8
- ☐ 5
- ☐ 7

Maximum marks: 2

6

Determining the convergence order

Referring back to the previous question, imagine that you wrote a computer program which implements the composite Boole's rule.

Describe in words how you would determine its convergence order experimentally.

Fill in your answer here

Maximum marks: 5

7 **Explicit midpoint rule**

Given the initial value problem

$y'(t) = \sqrt{y(t)}, \quad y(0) = 1$

Write down a **complete** implementation of the **explicit midpoint rule** (also known as the **improved explicit Euler method**) to find an approximation the solution to this initial value problem at $T = 2$.
NB! You can use either **Python or Matlab** syntax, but make sure that the program is complete and could be executed if you copied and pasted it into Matlab or a Jupyter notebook.
Fill in your answer here

1

Maximum marks: 10

8 **Determine convergence order**

Assume that you want to solve an initial value problem $y'(t) = f(t, y(t)), \quad y(t_0) = y_0$ numerically using a Runge-Kutta method which has a consistency error of $\mathcal{O}(\tau^5)$. What convergence order would you expect?

Select one alternative:

- ☐ 7
- ☐ 3
- ☐ 4
- ☐ 6
- ☐ 5

Maximum marks: 1

Theta method

The θ -method is a one-step method to solve initial value problems numerically and which for particular choices of $\theta \in [0, 1]$ reduces to the following schemes

Select one or more alternatives

- ☐ Explicit midpoint rule method
- ☐ Explicit Euler method
- ☐ Implicit Euler method
- ☐ Crank-Nicolson method
- ☐ Heun's method

In particular for $\theta = 0$ we obtain

Select one alternative

- ☐ Implicit Euler method
- ☐ Explicit midpoint rule
- ☐ Heun's method
- ☐ Explicit Euler method
- ☐ Crank-Nicolson method

For $\theta = 1$ we obtain

Select one alternative

- ☐ Implicit Euler method
- ☐ Explicit Euler method
- ☐ Explicit midpoint rule
- ☐ Heun's method
- ☐ Crank-Nicolsen method

For $\theta = \frac{1}{2}$ we obtain

Select one alternative

- ☐ Explicit midpoint rule
- ☐ Explicit Euler method
- ☐ Crank-Nicolson method
- ☐ Heun's method
- ☐ Implicit Euler method